

#Aviation Meteorology



Chapter Five: Thunderstorms



Objectives

- To Be aware of the Stages of Thunderstorms
- To Know atmospheric electricity and its Effect on Aviation operations
- To Identify key technologies used in aviation for thunderstorm avoidance
- To Explain the Effects of Downbursts in Aviation

- Thunderstorms are complex meteorological phenomena characterized by intense convective activity, producing thunder, lightning, heavy rainfall, and sometimes severe weather such as hail or tornadoes.
- Thunderstorms can occur as isolated cells, in clusters, or organized into larger systems like squall lines.

Structure of Thunderstorms

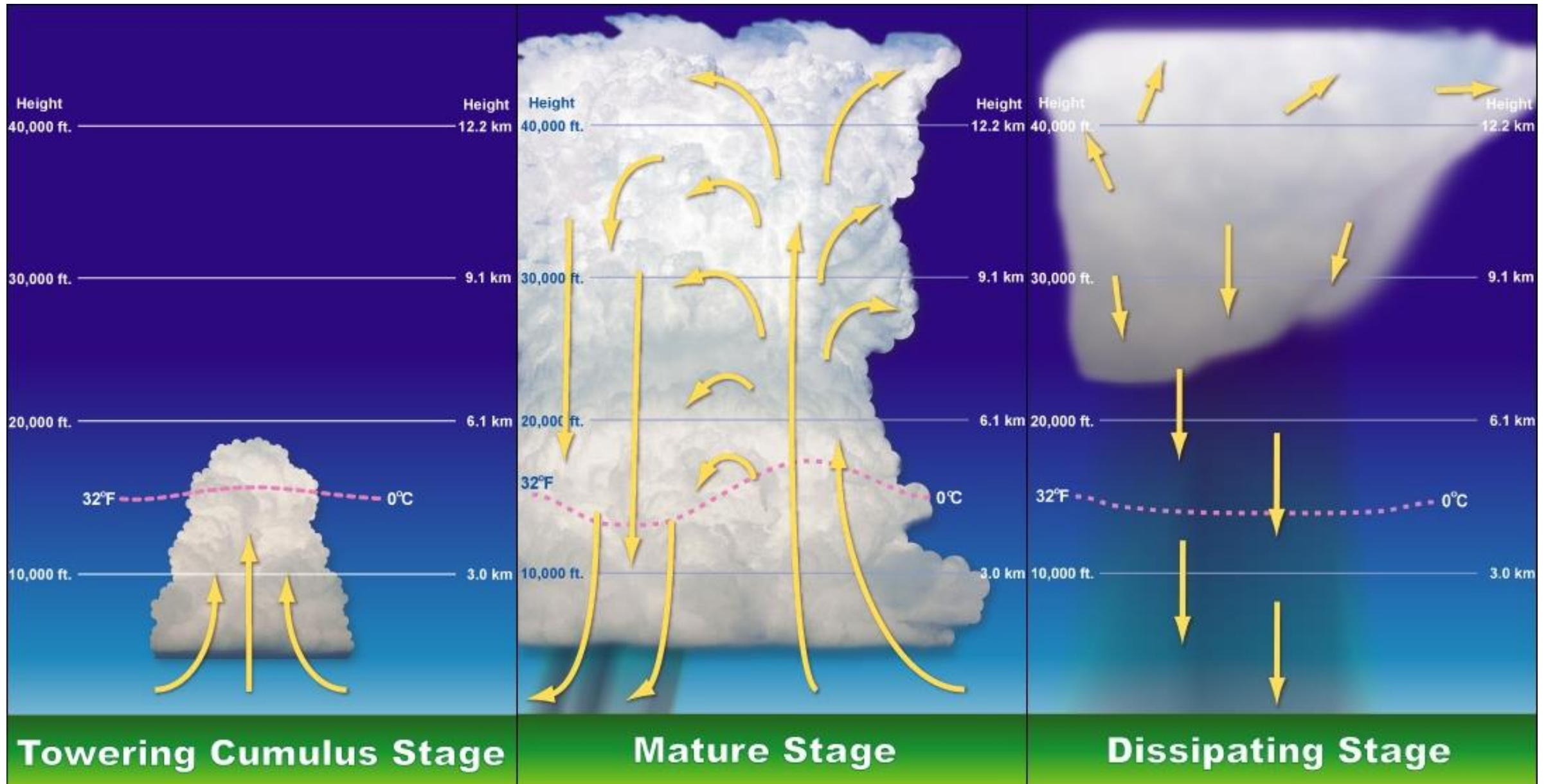
The Three main stages of a thunderstorms are

- *Developing stage*
- *Mature stage*
- *Dissipating stage*

- In the developing stage, rising air called “*updrafts*” are dominant in this stage and *towering cumulus clouds form*.
- This type of cloud is also known as a cumulus congestus.
- Next is the mature stage where the towering cumulus becomes a *cumulonimbus cloud* featuring both updrafts and “downdrafts” or sinking air.

- Downdrafts develop as a result of water droplets falling to the ground as precipitation (e.g., snow, rain, hail).
- The falling drops drag air downward as they fall, creating a downward current of air in the cumulonimbus cloud.
- Finally, in the *dissipating stage* downdrafts are dominant and cut off the updrafts needed for a cumulonimbus cloud to form and sustain itself. As a result, the cloud dissipates.

- If a cumulonimbus cloud has thunder and lightning, it is called a *thunderstorm*. Lightning is caused by a charge separation in clouds. For this charge separation to occur, a cloud must have ice crystals inside of it, which is why thunderstorms are always mixed-phase or ice-phase clouds.

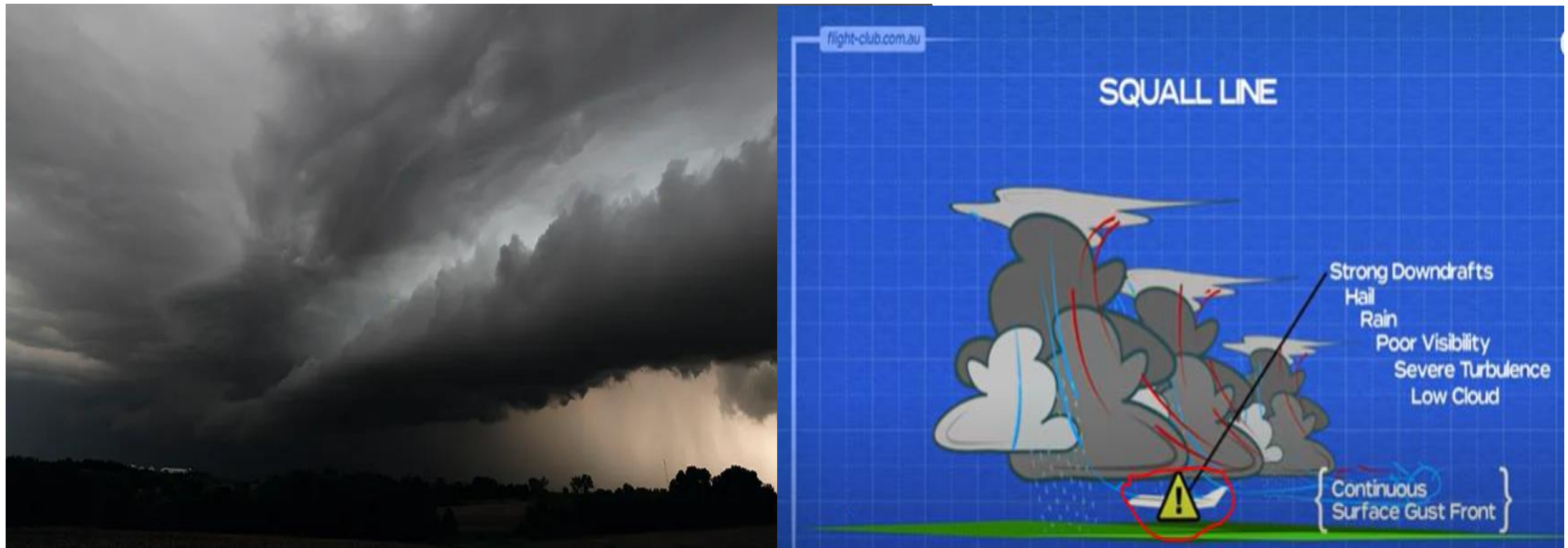


Severe Thunderstorm Types

Squall Line

- A squall line is a linear propagating severe thunderstorm.
- Squall lines consist of a long, narrow line of merged thunderstorms, often triggered by a cold front, dry line, or gust front.
- Squall lines can be many hundreds of kilometers long, but their width is usually much shorter, between 15 and 100 km.
- They can sustain themselves from several hours to several days.

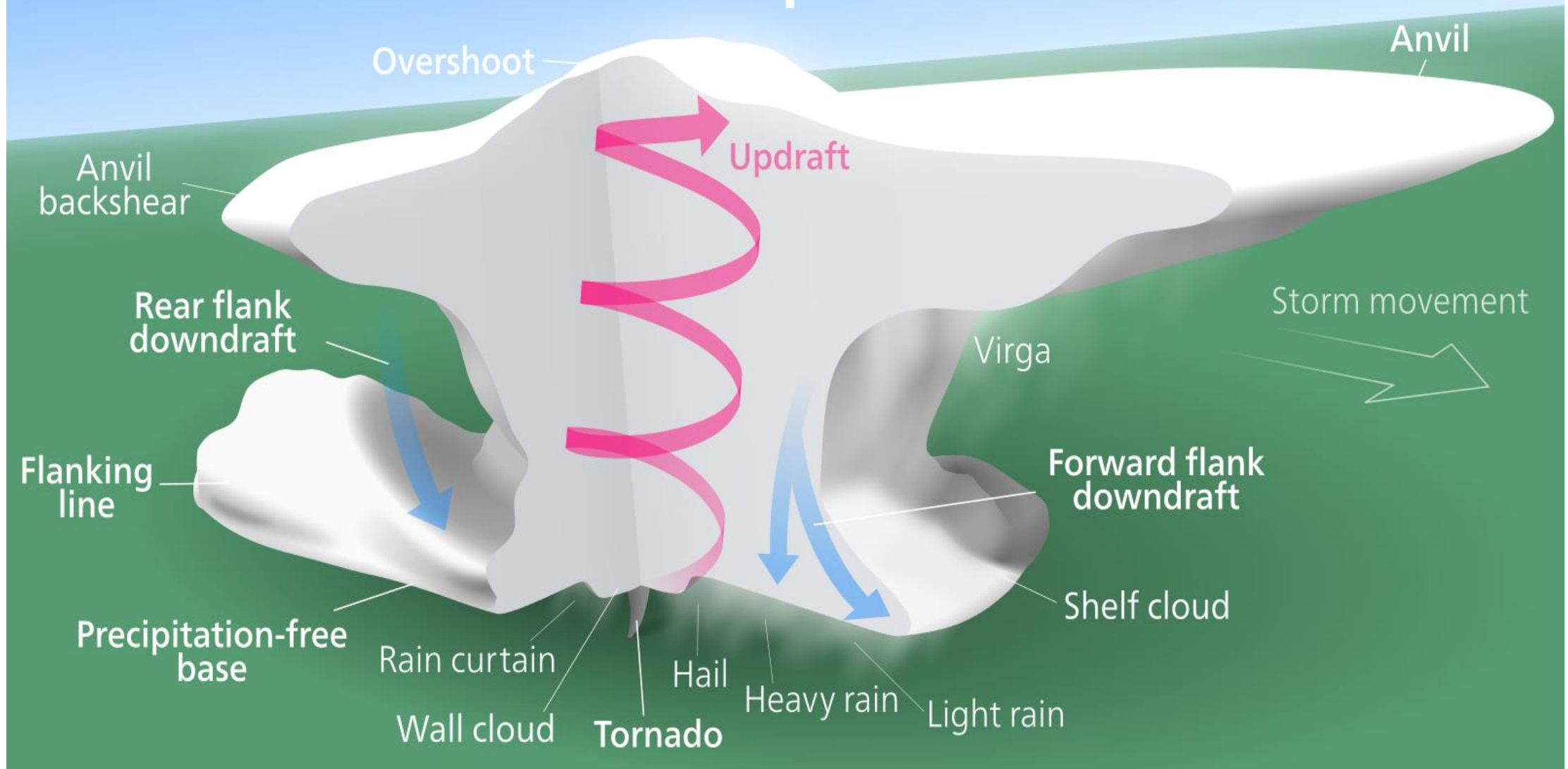
What will happen for flight in Squall Line?



Supercell

- A supercell is a special type of severe storm that has *rotation*.
- Supercells form when the environment has directional wind shear causing updrafts within a cloud to gain vorticity as they rise.

structure of a supercell



Electricity in the atmosphere

- Electricity in the atmosphere is primarily associated with atmospheric electricity, *a natural phenomenon that involves the presence and flow of electric charge in the Earth's atmosphere.*
- This phenomenon has several manifestations, including *lightning, thunderstorms, and the existence of electric fields* in the atmosphere.

- In aviation, understanding and managing the impact of electricity in the atmosphere is crucial for the safety and operation of aircraft. Here are some key considerations related to atmospheric electricity in aviation:

Lightning Hazards

- Lightning poses a significant hazard to aircraft.
- Lightning strikes can occur during thunderstorms, and they can potentially *damage aircraft systems*, including avionics and navigation equipment.
- To mitigate this risk, modern aircraft are equipped with lightning protection systems and conductive materials to dissipate electrical charges safely.

Static Electricity:

- Aircraft flying through the atmosphere can accumulate static electricity, particularly *during flight through clouds or precipitation*.
- Static discharges can occur during *fueling, loading, or unloading of passengers and cargo*. Proper grounding procedures are essential to prevent static-related issues.

Navigation and Communication:

- Atmospheric electricity can affect radio wave propagation, potentially leading to disruptions in communication and navigation systems.
- Pilots and air traffic controllers are trained to handle such situations and may use alternative frequencies or communication methods when needed.

- In summary, atmospheric electricity plays a significant role in aviation safety.
- Aircraft and systems are designed and operated with a deep understanding of these phenomena to ensure the safety and reliability of air travel.

Thunderstorm avoidance

- Thunderstorm avoidance is a critical aspect of aviation safety, and various technologies are employed to help pilots navigate and make informed decisions when encountering or anticipating thunderstorms.
- Here are some key technologies used in aviation for thunderstorm avoidance

Weather Radar (Airborne):

- Aircraft are equipped with weather radar systems that allow pilots to detect and navigate around areas of intense precipitation, including thunderstorms.
- Modern weather radars provide information about the location, intensity, and movement of storms. Pilots use this data to deviate from hazardous weather and find paths of least resistance.

Onboard Weather Radar Displays:

- Cockpit displays show the radar returns in a color-coded format, with different colors representing different levels of precipitation intensity.
- Pilots can use this information to identify areas of turbulence and avoid potential thunderstorm cells.

Stormscope:

- Stormscope is a weather detection system that identifies electrical discharges within thunderstorms, including lightning.
- It helps pilots visualize the extent and intensity of thunderstorm activity.
- Stormscope can provide a real-time display of lightning strikes relative to the aircraft's position, allowing pilots to make more informed decisions about route changes.

- It's important to note that thunderstorm avoidance is a dynamic process, and pilots continuously assess weather conditions during flight.
- Utilizing a combination of these technologies and receiving timely and accurate weather information helps ensure the safety of aircraft and passengers when operating in areas prone to thunderstorms.

Development and effect of downbursts

- Downbursts are powerful and rapidly descending air currents associated with thunderstorms.
- They can have significant effects on aviation, and understanding their development and impact is crucial for pilot safety.
- ***Formation:*** Downbursts typically form within thunderstorms when there is a rapid column of air descending from a cloud.
- This air column can be influenced by the presence of strong updrafts and downdrafts within the storm.

Effects of Downbursts in Aviation

- Wind Shear
- Turbulence
- Impact on Landing and Takeoff
- Reduced Lift

Effect of Lightning Strike on Aircraft and Flight Execution

- *The effects of a lightning strike on an aircraft can vary depending on several factors, including the severity of the strike, the specific systems affected, and the design features of the aircraft. Here are some potential effects:*

❖ ***Avionics Malfunctions*** Avionics, which include communication and navigation equipment, are susceptible to damage from electrical surges caused by lightning. Malfunctions in these systems could impact the pilot's ability to communicate or navigate.

❖ ***Electrical System Damage***

❖ ***Static Discharge***

❖ ***Fire Risk***

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Thanks !

#Aviation Meteorology



Chapter Six: Aircraft Icing



Objectives

- To Define aircraft icing and explain its effect on aviation operation
- To Know the formation of Aircraft icing
- To Describe the effect of Aircraft icing on aviation operation
- To Identify Different Types of Icing
- To Explain the Ice accretion rates

Aircraft Icing



- Aircraft icing refers to the *accumulation of ice on the surface of an aircraft during flight*.
- This can happen when the aircraft flies through clouds that contain supercooled water droplets or when flying in cold and humid conditions, such as during winter or at high altitudes.
- The accumulation of ice on an aircraft can affect its performance and handling characteristics and can lead to loss of control or even a crash.

- The ice adds *weight to the aircraft and changes its shape*, which can reduce lift and increase drag, leading to slower speeds, reduced maneuverability, and longer takeoff and landing distances.
- The formation of icing on an aircraft's surface is influenced by several factors, *including temperature, drop size, liquid water content, Airframe configuration, and Aircraft speed*

- The process begins when the aircraft enters a cloud containing supercooled water droplets.
- These droplets remain in a liquid state below the freezing point of water due to a lack of nucleation sites, which are necessary for ice formation.

❑ **Temperature:** The temperature of the aircraft's surface is a critical factor in the formation of icing.

- ✓ If the surface temperature is below freezing, any liquid water droplets that come into contact with it will rapidly freeze, forming a layer of ice.
- ✓ Lower temperatures lead to faster ice formation, as the supercooled droplets freeze more quickly on contact with the aircraft's surface.
- ✓ Therefore, it is essential to monitor the surface temperature of the aircraft, particularly when flying through areas with known icing conditions.

❑ **Drop size:** The size of the supercooled water droplets in the cloud can also have an impact on the formation of icing.

- ✓ Larger droplets tend to **freeze more quickly** than smaller droplets, as they contain more liquid water and have a higher surface area to volume ratio.
- ✓ This means that as the aircraft flies through a cloud containing **larger droplets**, it is more likely to accumulate ice more quickly than when flying through a cloud with **smaller droplets**.

❑ **Liquid water content:** The liquid water content (LWC) of a cloud refers to the amount of liquid water droplets present in the cloud.

- ✓ **Higher LWC** can lead to **more rapid** ice formation, as there are more droplets available to freeze on contact with the aircraft.
- ✓ Therefore, it is important to monitor the LWC of a cloud when flying through areas with known icing conditions.

❑ **Airframe configuration:** The design of an aircraft's airframe can impact the formation of icing.

- ✓ Some airframe **designs** are **more prone** to ice formation than others.
- ✓ For example, airfoils with a high camber or thickness can create areas of low pressure that promote the formation of ice.
- ✓ Therefore, it is important to consider the airframe configuration when **designing an aircraft** and when flying through areas with known icing conditions.

❑ **Aircraft speed:** The speed of the aircraft can also impact the formation of icing.

- ✓ Flying at **high speeds** can cause the supercooled droplets to shatter on impact with the aircraft's surface, which can lead to less ice accumulation.
- ✓ However, flying **too fast** can also cause the ice to build up more quickly due to the higher air pressure and cooling effect at high speeds.
- ✓ Therefore, pilots must consider the **appropriate speed** for their aircraft when flying through areas with known icing conditions.

- In conclusion, the formation of icing on an aircraft's surface is influenced by several factors, including *temperature, drop size, liquid water content, airframe configuration and aircraft speed.*
- By monitoring these factors, pilots and aircraft maintenance personnel can take steps to mitigate the effects of icing and ensure safe flight operations.
- This may involve avoiding areas with known icing conditions, using **deicing** and **anti-icing** systems, and maintaining a safe airspeed and altitude.

Types of Aircraft Icing

❖ There are three main types of aircraft icing:

- ✓ Structural icing

- ✓ Induction icing

- ✓ Instrument icing

❖ **Structural icing:** Structural icing occurs when ice accumulates on the *external surfaces of an aircraft*, including the wings, fuselage, and tail.

- ✓ This type of icing can occur in any weather condition where the temperature is below freezing and there is moisture in the air, including clouds, fog, and precipitation.
- ✓ As ice accumulates on the aircraft's surfaces, it can change their *shape and reduce their efficiency*, leading to reduced lift and increased drag.

- ✓ This can result in slower speeds, reduced maneuverability, and longer takeoff and landing distances.
- ✓ Structural icing can be further divided into clear ice, rime ice, and mixture of two.

❖ **Induction icing:** Induction icing occurs when ice accumulates on the *engine's air intake*, disrupting the **flow of air** to the engine and reducing its performance.

- ✓ Induction icing is particularly **dangerous** as it can lead to a **loss of engine power** and possible engine **failure**.
- ✓ Induction icing can occur in conditions where the temperature is **near** or **below freezing** and there is **moisture** in the air.

❖ **Instrument icing:** Instrument icing occurs when ice accumulates on the *aircraft's instruments*, such as the **pitot tubes** and **static ports**.

- ✓ This can cause **incorrect readings** and a **loss** of important information for the pilot, leading to potential control issues.
- ✓ Instrument icing can occur in conditions where the temperature is **near or below freezing** and there is **moisture** in the air.

Ice accretion rates

- Ice accretion rates refer to the **rate** at which ice **accumulates** on an aircraft's surfaces.
- the rate of ice accretion on an aircraft's surfaces is influenced by several factors, including **cloud type, thunderstorms, freezing precipitation, and orographic and frontal lifting** effects.

Ice accretion rates

a) Cloud types: The type of cloud can affect the rate of ice accretion.

- ✓ **Stratiform clouds**, which are flat and layered, typically have a **lower** LWC and produce **smaller** droplets, resulting in a **slower** rate of ice accretion.
- ✓ Cumuliform clouds, which are tall and fluffy, have a **higher** LWC and produce **larger droplets**, resulting in a **faster** rate of ice accretion.
- ✓ This is because cumuliform clouds have more vertical motion, which allows for more mixing of the supercooled droplets.

Ice accretion rates

b) Thunderstorms: Thunderstorms can produce significant amounts of **supercooled water droplets, hail**, and other **frozen particles** that can quickly accumulate on an aircraft's surfaces.

✓ The **rate of ice accretion** during a thunderstorm can be much **higher** than in other **cloud types**, posing a significant **hazard** to aircraft.

Ice accretion rates

c) Freezing precipitation: Freezing precipitation, such as **freezing rain** or **sleet**, can also lead to **rapid** ice accretion on an aircraft's surfaces.

- ✓ When supercooled droplets come into contact with the aircraft, they can freeze on impact and quickly accumulate, leading to **dangerous** icing conditions.

Ice accretion rates

d) Orographic and frontal lifting effects: Orographic lifting occurs when air is forced to **rise** over a mountain or other topographic feature, leading to **cloud formation** and potential **icing** conditions.

- ✓ Frontal lifting occurs when **warm** and **cold** air masses **meet**, leading to the formation of clouds and potential icing conditions.
- ✓ These lifting effects can **increase** the **LWC** and the **rate of ice accretion**, leading to more **dangerous** icing conditions.

Ice accretion rates

- In conclusion, the rate of ice accretion on an aircraft's surfaces is influenced by several factors, including cloud type, thunderstorms, freezing precipitation, and orographic and frontal lifting effects.
- **Pilots and aircraft maintenance personnel** must be **aware** of these factors and **take appropriate** measures to **prevent** or **mitigate** the effects of icing to ensure safe flight operations.

Deicing and Anti-Icing Equipment

- Many aircraft have **some, but not all**, the **gear** required for approved flight into icing conditions.
- In some cases, the equipment has been **added** as an after-market modification.

Deicing and Anti-Icing Equipment

➤ **Anti-Icing equipment** is turned on **before** the flight enters icing conditions.

✓ Typically this includes carburetor heat, prop heat, pitot heat, fuel vent heat, windshield heat, and fluid surface deicers (in some cases).

➤ **Deicing** is used **after** ice has built up to an appreciable amount.

✓ Typically this includes **surface deice** equipment.

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Thanks !